

Power BI Basics

Question 1 : What is Power BI and why is it used in businesses?

Answer:- Power BI is Microsoft's business intelligence tool that helps organizations turn raw data into interactive dashboards and reports, enabling faster, more accurate decision-making. Businesses use it to integrate data from multiple sources, visualize trends, and improve efficiency across finance, sales, operations, and more.

- **Definition:** Power BI is a cloud-based business intelligence and analytics platform developed by Microsoft.
- **Core Function:** It allows users to connect to various data sources (Excel, SQL databases, cloud services, CRMs, ERPs), clean and transform data, and create interactive visualizations.
- **Accessibility:** Available as a desktop app (Power BI Desktop), online service (Power BI Service), and mobile app.

Businesses Use Power BI

1. Data Integration

- Combines data from **multiple sources** (Excel, SQL Server, Salesforce, Google Analytics, etc.) into one unified view.
- Eliminates manual reporting and reduces errors.

2. Real-Time Dashboards

- Dashboards update automatically as data changes.
- Managers can monitor KPIs (sales, revenue, inventory, customer satisfaction) in real time.

3. Decision-Making Support

- Provides **interactive charts, graphs, and maps** to spot trends and anomalies.
- Helps executives and teams make **data-driven decisions** instead of relying on intuition.

4. Collaboration

- Reports and dashboards can be shared across teams via **Microsoft Teams, Outlook, or mobile apps**.
- Supports comments, alerts, and subscriptions for ongoing monitoring.

5. Cost & Efficiency

- Saves time compared to manual Excel reporting.
- Many organizations report **break-even within a year** due to productivity gains.

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Real-World Business Use Cases

Function	Example Use Case	Benefit
Finance	Automated P&L and balance sheets	Faster monthly closes, fewer errors
Sales	Track performance by region/product	Real-time quota monitoring, predictive insights
Supply Chain	Inventory tracking across locations	Prevents stockouts/overstocking, reduces costs
Education	Student retention dashboards	Identifies at-risk students early
Banking	Fraud detection	Detects anomalies in minutes

Risks & Considerations

- **Data Quality Dependency:** Poor or inconsistent data leads to misleading dashboards.
- **Learning Curve:** Requires training for effective use.
- **Licensing Costs:** Power BI Pro or Premium subscriptions may be needed for enterprise-scale sharing.
- **Security:** Sensitive data must be managed carefully with role-based access.

Question 2 : Name and explain the three main components of Power BI.

Answer:- The three main components of Power BI are Power Query, Power Pivot, and Power View. Together, they handle data preparation, modeling, and visualization—forming the backbone of how Power BI transforms raw data into actionable insights.

Three Main Components of Power BI

1. Power Query (Data Preparation)

- **Purpose:** Used to connect, import, and transform data from multiple sources (Excel, SQL Server, web pages, cloud services, etc.).
- **Functions:**
 - Clean messy datasets (remove duplicates, rename columns, filter rows).
 - Merge or append tables from different sources.
 - Perform “data wrangling” without coding, using a visual interface.
- **Benefit:** Ensures that only accurate, structured data flows into reports, reducing errors and manual effort.

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2. Power Pivot (Data Modeling & Calculations)

- **Purpose:** Acts as the analytical engine behind Power BI, enabling relationships and calculations across large datasets.
- **Functions:**
 - Create **data models** by linking multiple tables.
 - Use **DAX (Data Analysis Expressions)** formulas for advanced calculations (e.g., profit margins, growth rates).
 - Handle millions of rows efficiently with compression technology.
- **Benefit:** Allows businesses to perform complex analysis quickly, even on massive datasets, without slowing down performance.

3. Power View (Data Visualization)

- **Purpose:** Provides interactive charts, graphs, and maps to visualize data.
- **Functions:**
 - Drag-and-drop fields to create visuals like bar charts, line charts, pie charts, and geographic maps.
 - Interactive filtering—clicking on one chart updates others instantly.
 - Helps spot trends, anomalies, and patterns visually.
- **Benefit:** Makes insights accessible to non-technical users, enabling decision-makers to understand data at a glance.

They Work Together

- **Power Query** → Cleans and prepares the data.
- **Power Pivot** → Models and calculates relationships.
- **Power View** → Visualizes the results in dashboards and reports.

This workflow ensures that businesses can go from raw data to actionable insights seamlessly.

Risks & Considerations

- **Data Quality:** If Power Query doesn't clean data properly, Power Pivot and Power View will produce misleading results.
- **Learning Curve:** DAX formulas in Power Pivot require practice.
- **Integration Needs:** For enterprise use, combining with Power BI Service (cloud sharing) is essential.

Question 3 : Explain the Power BI workflow.

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Answer :- Power BI's workflow is a step-by-step process that takes raw, messy data and transforms it into clean, modeled, and visualized insights that businesses can use for decision-making. It typically involves data collection, cleaning, modeling, visualization, and sharing.

Power BI Workflow Steps

1. Data Collection

- Connect to multiple sources: **Excel, CSV, SQL databases, APIs, cloud services.**
- Choose between **Import Mode** (fast, data stored in Power BI) or **DirectQuery Mode** (real-time queries from source).
- At this stage, data is often incomplete or inconsistent.

2. Data Cleaning & Transformation (Power Query)

- Use **Power Query Editor** to prepare data.
- Common tasks:
 - Remove duplicates
 - Handle missing values
 - Change data types
 - Merge or split columns
- Every step is recorded, making transformations repeatable and transparent.
- **Key risk:** Skipping cleaning leads to misleading dashboards later.

3. Data Modeling (Power Pivot)

- Define relationships between tables (e.g., Sales ↔ Products ↔ Customers).
- Create hierarchies and manage filters.
- Use **DAX (Data Analysis Expressions)** for calculations like profit margins, growth rates, or KPIs.
- **Benefit:** A strong model ensures accuracy and prevents broken slicers or incorrect totals.

4. Calculations with DAX

- Build measures and calculated columns.
- Examples:
 - Year-to-Date Sales
 - Customer Retention Rate
 - Average Order Value
- DAX enables advanced analytics beyond simple sums or averages.

5. Visualization & Reporting (Power View)

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- Create interactive dashboards with charts, maps, KPIs, and slicers.
- Features: drill-through, conditional formatting, forecasting, and trend lines.
- Reports are dynamic—clicking one chart updates others instantly.

6. Sharing & Collaboration (Power BI Service)

- Publish dashboards to the **Power BI Service (cloud)**.
- Share securely with teams via **Microsoft Teams, Outlook, or mobile apps**.
- Set up **scheduled refreshes** to keep reports up to date.
- Role-based access ensures data security.

Workflow Summary Table

Step	Tool/Feature	Purpose	Risk if Skipped
Data Collection	Import/DirectQuery	Connect to sources	Missing/slow data
Data Cleaning	Power Query	Prepare clean dataset	Misleading insights
Data Modeling	Power Pivot	Relationships & structure	Broken filters, wrong totals
Calculations	DAX	Advanced measures	Limited analysis
Visualization	Power View	Interactive dashboards	Poor decision support
Sharing	Power BI Service	Collaboration & refresh	Data silos, outdated reports

Risks & Considerations

- **Data Quality Dependency:** Garbage in → garbage out.
- **Performance Issues:** Poor modeling or wrong connection mode slows reports.
- **Security:** Sensitive data must be shared with role-based permissions.

Question 4 : List any four data cleaning tasks that can be performed in Power Query.
Answer:- Four common data cleaning tasks we can perform in Power Query within Power BI:

Four Data Cleaning Tasks in Power Query

1. Remove Duplicates

- **What it does:** Identifies and deletes repeated rows in a dataset.

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- **Example:** If a sales transaction is recorded twice, Power Query can remove the duplicate entry to avoid inflated revenue figures.

2. Handle Missing Values

- **What it does:** Deals with null or blank fields.
- **Techniques:** Replace with default values, fill down/up, or remove incomplete rows.
- **Example:** Filling missing “Customer Name” with “Unknown” or replacing blank exam scores with the class average.

3. Change Data Types

- **What it does:** Ensures columns have the correct format (text, number, date, currency).
- **Example:** Converting a column stored as text (“2026-06-23”) into a proper Date type for time-based analysis.

4. Split or Merge Columns

- **What it does:** Breaks down or combines fields for consistency.
- **Example:** Splitting “Full Name” into “First Name” and “Last Name,” or merging “City” and “State” into a single “Location” column.

Question 5 : Write step by step instructions to load the above dataset.

Answer:- **Real_Estate_Data_100_Rows.csv** dataset load-ready in Power BI, you need to follow a structured workflow. Here’s a clear step-by-step guide:

01

Open Power BI Desktop

Start by launching Power BI Desktop to begin the data loading process.

- Ensure you have Power BI Desktop installed
- Open the application from your system

02

Connect to the CSV File

Import the dataset from your local storage.

Home → Get Data → Text/CSV

- Click **Get Data** on the ribbon

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- Select **Text/CSV**
- Browse and choose `Real_Estate_Data_100_Rows.csv`
- Click **Load** to preview data

03

Review Data in Power Query

Critical Step

Check for missing values, duplicates, and formatting issues before loading.

- Identify blank fields (e.g., missing Price, Owner_Name, Contact)
- Detect duplicate Property_ID entries (e.g., ID 110 appears twice)
- Verify data types (Price as number, Area_sqft as number, Contact as text)

04

Apply Data Cleaning

Transform the dataset to ensure consistency and accuracy.

- Remove duplicate rows
- Replace null values with defaults or averages
- Correct data types (e.g., Price → Decimal Number)
- Standardize text fields (City, Location_Details)

05

Load Cleaned Data into Model

Push the cleaned dataset into Power BI for analysis.

- Click **Close & Load** in Power Query
- Data is now available in Power BI's data model
- Begin creating relationships, measures, and dashboards

Question 6 : Define Data View, Report View, and Model View. Explain the purpose of each view.

Answer:- In Power BI, we work with three main **views**—each serving a different purpose in the workflow of building reports and dashboards. Here's a clear breakdown:

1. Data View

- **Purpose:** Displays the actual data tables after they've been loaded and transformed.

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- **What you see:** Rows and columns similar to Excel, showing the cleaned dataset.
- **Use case:**
 - Verify that data has been imported correctly.
 - Check transformations (e.g., missing values handled, duplicates removed).
 - Inspect calculated columns created with DAX.

2. Report View

- **Purpose:** The main canvas for creating **visualizations and dashboards**.
- **What you see:** Charts, graphs, maps, slicers, KPIs, and interactive visuals.
- **Use case:**
 - Build reports by dragging fields onto visuals.
 - Apply filters and slicers for interactivity.
 - Design dashboards for business users to explore insights.

3. Model View

- **Purpose:** Defines the **relationships** between tables and manages the data model.
- **What you see:** A diagram of tables connected by relationships (like ER diagrams in databases).
- **Use case:**
 - Create relationships (one-to-many, many-to-many).
 - Manage hierarchies (e.g., Year → Quarter → Month).
 - Ensure filters and slicers work correctly across multiple tables.

They Work Together

- **Data View** → Inspect and validate the dataset.
- **Model View** → Build relationships and structure the data model.
- **Report View** → Create interactive dashboards and share insights.

Question 7 : Discuss the different data sources that Power BI supports.

Answer:- Power BI supports a wide range of data sources, including files (Excel, CSV, JSON), databases (SQL Server, Oracle, MySQL), cloud services (Azure, Google BigQuery), and online platforms (Salesforce, Facebook, Google Analytics). This flexibility allows businesses to integrate diverse datasets into one unified reporting environment.

Categories of Data Sources in Power BI

1. File-Based Sources

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- **Excel Workbook (.xlsx)** – Most common starting point for reports.
- **Text/CSV (.txt, .csv)** – Simple tabular data import.
- **XML & JSON** – Structured data formats often used in APIs.
- **PDF Documents** – Extract tabular data from PDFs.
- **Folders** – Combine multiple files from a directory into one dataset.
- **SharePoint Folder** – Import files stored in SharePoint.

2. Database Sources

- **SQL Server** (including Analysis Services)
- **Oracle Database**
- **MySQL, PostgreSQL**
- **IBM DB2, Sybase, Teradata**
- **Access Database**
- **SAP HANA, SAP BW**
- **Snowflake, Amazon Redshift**

These connections can use **Import Mode** (data copied into Power BI) or **DirectQuery Mode** (live connection to the database).

3. Cloud-Based Sources (Azure & Others)

- Azure SQL Database
- Azure Synapse Analytics
- Azure Analysis Services
- Azure Blob Storage / Data Lake
- Google BigQuery
- Amazon Web Services (AWS) sources

4. Online Services

- Salesforce Objects & Reports
- Google Analytics
- Facebook & LinkedIn data
- Dynamics 365
- Exchange Online
- Active Directory

5. Other Sources

- **ODBC Connections** – Connect to any ODBC-compliant source.
- **Hadoop (HDFS)** – For big data environments.
- **R Scripts / Python Scripts** – Import data generated by custom code.
- **Web Data** – Scrape tables directly from websites.

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This Matters

- **Flexibility:** Businesses can combine structured (databases) and unstructured (JSON, web data) sources.
- **Scalability:** Cloud connectors like Azure Synapse or BigQuery handle massive datasets.
- **Real-Time Insights:** DirectQuery enables live dashboards without scheduled refreshes.

Question 8 : Split Owner Name to create two new columns as First Name and Last Name

Answer:- To split the **Owner_Name** column into **First Name** and **Last Name** in Power BI using **Power Query**, follow these step-by-step instructions:

Steps in Power Query

1. **Load the Dataset**
 - Open Power BI Desktop.
 - Go to **Home** → **Get Data** → **Text/CSV**.
 - Select your file **Real_Estate_Data_100_Rows.csv** and click **Load**.
2. **Open Power Query Editor**
 - In Power BI, click **Transform Data**.
 - This opens the **Power Query Editor** where you can clean and transform data.
3. **Select the Owner_Name Column**
 - Locate the column **Owner_Name** in your dataset.
 - Highlight it by clicking on the column header.
4. **Split Column by Delimiter**
 - Go to the **Home** or **Transform** tab.
 - Click **Split Column** → **By Delimiter**.
 - Choose **Space** as the delimiter (since names are separated by a space).
 - Select **Split into Two Columns**.
5. **Rename the Columns**
 - The split will create two new columns: **Owner_Name.1** and **Owner_Name.2**.
 - Rename them to:
 - i. **First Name**
 - ii. **Last Name**
6. **Handle Missing Values**
 - Some rows have blank Owner_Name (e.g., Palm Residency).
 - For those, the split will produce null values.
 - You can replace nulls with **"Unknown"** or leave them blank depending on your reporting needs.
7. **Close & Apply**
 - Once done, click **Close & Apply** to load the transformed dataset back into Power BI.

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Example Result

Property_Name	Owner_Name	First Name	Last Name
Green Heights	Rahul Mehta	Rahul	Mehta
Sunrise Apartments	Anita Sharma	Anita	Sharma
Palm Residency	(blank)	Unknown	Unknown
